Object Detection for Dummies Part 2: CNN, DPM and Overfeat

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1 Object Detection for Dummies Part 2: CNN, DPM and Overfeat

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Part 1 of the "Object Detection for Dummies" series introduced: (1) the concept of image gradient vector and how HOG algorithm summarizes the information across all the gradient vectors in one image; (2) how the image segmentation algorithm works to detect regions that potentially contain objects; (3) how the Selective Search algorithm refines the outcomes of image segmentation for better region proposal.

In Part 2, we are about to find out more on the classic convolution neural network architectures for image classification. They lay the ***foundation*** for further progress on the deep learning models for object detection. Go check Part 3 if you want to learn more on R-CNN and related models.

Links to all the posts in the series: [Part 1] [Part 2] [Part 3] [Part 4].

convolution-operation.png

Figure 1: An illustration of applying a kernel on the input feature map to generate the output. (Image source: River Trail documentation)

2 CNN for Image Classification#

CNN, short for "**Convolutional Neural Network**", is the go-to solution for computer vision problems in the deep learning world. It was, to some extent, inspired by how human visual cortex system works.

2.1 Convolution Operation#

I strongly recommend this guide to convolution arithmetic, which provides a clean and solid explanation with tons of visualizations and examples. Here let's focus on two-dimensional convolution as we are working with images in this post.

In short, convolution operation slides a predefined kernel (also called "filter") on top of the input feature map (matrix of image pixels), multiplying and adding the values of the kernel and partial input features to generate the output. The values form an output matrix, as usually, the kernel is much smaller than the input image.

Figure 2 showcases two real examples of how to convolve a 3x3 kernel over a 5x5 2D matrix of numeric values to generate a 3x3 matrix. By controlling the padding size and the stride length, we can generate an output matrix of a certain size.

7 Citation

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e-2e-2 [t]1.956521739130435e-2

Figure 2: Two examples of 2D convolution operation: (top) no padding and 1x1 strides; (bottom) 1x1 border zeros padding and 2x2 strides. (Image source: deeplearning.net)

alex_net_illustration.p

e-2e-2 e-2e-2 [t]1.956521739130435e-2

2.2 AlexNet (Krizhevsky et al, 2012)#

- 5 convolution [+ optional max pooling] layers + 2 MLP layers + 1 LR layer
- Use data augmentation techniques to expand the training dataset, such as image transla-

8 References

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